Creating KML Marker Files in Google Earth

This section explains how to create simple KML marker files in Google Earth Pro. KML files are useful for sharing route reference points, station locations, landmarks, survey notes, and other geographic markers.

KML and KMZ

Google Earth can save placemarks and folders in two related formats:

KML

A plain text XML file. This is the best choice when the file should be readable, editable, documented, or generated by scripts.

KMZ

A zipped KML package. This is useful when the marker set includes bundled icons, images, overlays, or other resources.

For simple marker lists, use KML.

Creating a Single Marker

1. Open Google Earth Pro.
2. Navigate to the location where the marker should be placed.
3. Click the yellow pushpin button, or choose menu: Add[Placemark].
4. Enter a name for the marker.
5. Optionally enter a description.
6. Position the marker:
    a. Drag the pushpin to the exact location, or
    b. Enter latitude and longitude manually in the placemark dialog.
7. Click btn:[OK].
8. In the Places panel, right-click the new placemark.
9. Choose menu: Save Place As[].
10. In the file type dropdown, choose KML instead of KMZ.
11. Save the file.

Creating Multiple Markers

For a group of related markers, place them in a folder before saving.

1. In the Places panel, right-click My Places or Temporary Places.
2. Choose menu: Add[Folder].
3. Name the folder.
4. Add placemarks inside that folder.
5. Check that each placemark has the correct name, position, and description.
6. Right-click the folder.
7. Choose menu: Save Place As[].
8. Select KML as the output type.
9. Save the file.

Saving a folder creates one KML file containing all placemarks in that folder.

Coordinate Order

KML coordinates use this order:

```
longitude,latitude,elevation
```

This is different from the common latitude/longitude order used in many mapping tools.

For example:

```
<coordinates>-71.0589,42.3601,0</coordinates>
```

means:

Value	Meaning
`-71.0589`	Longitude
`42.3601`	Latitude
`0`	Elevation

Basic KML Marker File Example

A simple KML marker file looks like this:

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<kml xmlns="http://www.opengis.net/kml/2.2">
  <Document>
    <name>Route Markers</name>

    <Placemark>
      <name>Station A</name>
      <description>Example station marker</description>
      <Point>
        <coordinates>-71.0589,42.3601,0</coordinates>
      </Point>
    </Placemark>

  </Document>
</kml>
```

Multiple Marker Example

```
<?xml version="1.0" encoding="UTF-8"?>
<kml xmlns="http://www.opengis.net/kml/2.2">
  <Document>
    <name>Route Reference Markers</name>

    <Placemark>
      <name>Station A</name>
      <description>Main station reference point</description>
      <Point>
        <coordinates>-71.0589,42.3601,0</coordinates>
      </Point>
    </Placemark>

    <Placemark>
      <name>Yard Entrance</name>
      <description>Approximate start of yard trackwork</description>
      <Point>
        <coordinates>-71.0610,42.3588,0</coordinates>
      </Point>
    </Placemark>

  </Document>
</kml>
```

Good Practices

- Use clear marker names.
- Add descriptions when the marker purpose is not obvious.
- Keep related markers in folders.

- Save simple marker sets as `.kml` so they remain readable.
- Use `.kmz` only when bundled icons, overlays, or images are needed.
- Check coordinates carefully before using the KML as a route-building reference.

WARNING	KML uses `longitude,latitude,elevation`. If coordinates are entered in `latitude,longitude` order, the marker may appear in the wrong part of the world.

Suggested Route-Building Uses

KML marker files can be useful for documenting:

- station locations
- mileposts
- yards and sidings
- bridge and tunnel locations
- road crossings
- signal reference points
- scenery landmarks
- survey notes
- route boundaries

For route-building documentation, a KML marker file provides a lightweight way to preserve geographic reference points outside the route editor.